

MATH3171 (Mathematical Biology III)
YEAR 2025–26, MICHAELMAS TERM

PROBLEM SHEET 1

This problem sheet is due in by **5pm** on **Wednesday 29 October 2025**. You may submit it on paper to my office (MCS 3008) or electronically on Gradescope. There is a discretionary mark available for neat and legible solutions.

Topics covered: Chapters 1–2.

1. Consider the following growth model for a population $x(t)$. Known as the Gompertz model, it is given by

$$\frac{dx}{dt} = ax \log \left(\frac{K}{x} \right), \quad (1)$$

where a and K are positive constants.

- (a) Plot the phase space of this model and use this plot to determine the stability of any equilibria you find.
- (b) Consider now the biological interpretation of this model. What do the terms in the equation represent? And how might you interpret how this model differs from the logistic model? You only need to write a sentence or two.

2. The logistic model for the evolution of a prey species with density $x(t)$ can be adapted to form

$$\frac{dx}{dt} = Rx \left(1 - \frac{x}{K} \right) - P, \quad (2)$$

where R , K and P are positive constants.

- (a) Provide a brief biological interpretation of this model.
- (b) There is a biological problem with this model at $x = 0$. What is it?

To fix this problem, consider now the more involved model,

$$\frac{dx}{dt} = Rx \left(1 - \frac{x}{K} \right) - P \left(1 - \exp \left[-\frac{x^2}{\alpha^2 A^2} \right] \right), \quad (3)$$

where R , K , P , A and α are positive constants.

- (c) The units of x are density, the units of t are time, and α is just a number. By considering the new model, eq. (3), what, therefore, do the units of R , K , P and A have to be?
- (d) By writing

$$x = \hat{x}X, \quad t = \hat{t}T, \quad (4)$$

where $\hat{x}$ and $\hat{t}$ are nondimensional, show that eq. (3) can be written as

$$\frac{d\hat{x}}{d\hat{t}} = r\hat{x} \left(1 - \frac{\hat{x}}{q} \right) - \left(1 - \exp \left[-\frac{\hat{x}^2}{\alpha^2} \right] \right), \quad (5)$$

where you should give the values of X , T , r and q in terms of the original parameters.

- (e) Assume now that $\alpha \ll q$. Sketch the phase space of the nondimensionalised model and show that the problem with original model in eq. (2) has been solved. It may help to first sketch the phase space of just the exponential term,

$$\frac{d\hat{x}}{d\hat{t}} = \exp \left[-\frac{\hat{x}^2}{\alpha^2} \right], \quad (6)$$

marking α roughly on the x -axis.

- (f) *Optional:* Continue to assume that $\alpha \ll q$. Demonstrate that in order for there to be three nonzero equilibria, we must have $rq > 4$.

(Hint: if you can only spot two possible nonzero equilibria, quickly calculate the slope of the phase space plot at $x = 0$ and see what that implies...)

3. The following equations represent a pair of interacting biological populations:

$$\frac{dx}{dt} = x - 2y + 4, \quad (7)$$

$$\frac{dy}{dt} = 2x + 2y - 10. \quad (8)$$

- (a) Discuss each term in the equations: what do they represent (growth? decay? interaction?). Are these meaningful equations? ('I do/don't think this is realistic because...')
- (b) Find the equilibrium of the system: state its permissibility and determine its stability. You do not need to do a formal ε -style expansion; you can simply state a useful result from lectures to start you off.
- (c) Take a look at table 3.1 in the online notes on Ultra. What name should we give to this equilibrium?